

3.5 Unit 5 BIOL5 Control in cells and in organisms

Multicellular organisms are able to control the activities of different tissues and organs within their bodies. They do this by detecting stimuli and stimulating appropriate effectors: plants use specific growth factors; animals use hormones, nerve impulses or a combination of both. By responding to internal and external stimuli, animals increase their chances of survival by avoiding harmful environments and by maintaining optimal conditions for their metabolism.

Cells are also able to control their metabolic activities by regulating the transcription and translation of their genome. Although the cells within an organism carry the same genetic code, they translate only

part of it. In multicellular organisms, this control of translation enables cells to have specialised functions, forming tissues and organs. The sequencing and manipulation of DNA has many medical and technological applications.

Consideration of control mechanisms underpins the content of this unit. Students who have studied it should develop an understanding of the ways in which organisms and cells control their activities. This should lead to an appreciation of common ailments resulting from a breakdown of these control mechanisms and the use of DNA technology in the diagnosis and treatment of human diseases.
